

# Evidence-Bound Curation: A Traceable Framework for Language-Model Training Data

Keonsoo Lee<sup>\*†</sup>, Jay Hoon Jung<sup>†</sup>, and Shaikh Arifuzzaman<sup>\*</sup>

<sup>\*</sup>Data-intensive Scalable Computing Lab, University of Nevada, Las Vegas (UNLV), Las Vegas, NV, USA

<sup>†</sup>Korea Aerospace University, Gyeonggi-do, South Korea

Emails: leek133@unlv.nevada.edu, jhjung@kau.ac.kr, shaikh.arifuzzaman@unlv.edu

**Abstract**—Modern data curation pipelines combine rules, similarity search, deduplication, and learned quality scores, but often make it difficult to trace why an individual example was kept, removed, quarantined, or restored. We introduce evidence-bound curation, a framework that separates evidence from authority: each curation signal emits inspectable evidence, including pass/fail/abstain judgments where applicable, and an explicit policy determines which evidence may authorize which action on the corpus. Rather than collapsing Validity, Redundancy, Quality, and Coverage into interchangeable keep/drop votes, the framework gives each an asymmetric decision role. Validity failures isolate an item directly. Redundancy may remove an item only when a supporting relation is verified and a representative is retained. Quality is assessed by a frozen four-head ranker whose heads return pass, fail, or abstain. The policy admits an item only when semantic coherence, substantive payload, and recoverable learnable relations provide positive support; correctness acts solely as a qualified-failure veto. Coverage may restore previously discarded items but can never initiate a deletion. Every accepted action records a content fingerprint, policy version, decision reason, and, where applicable, a representative or restoration link, making the corpus reproducible and auditable at the level of individual decisions. We instantiate one frozen policy on code and mathematics corpora and evaluate it by continued pretraining on all retained packed tokens, against no-update and Raw controls and same-corpus Data-Juicer and NeMo Curator recipes under a common protocol. Retaining 89.4% of code and 82.6% of math packed tokens, the frozen policy outperformed both curation baselines on three of six code benchmarks but trailed both baselines on all four math benchmarks. Auditable, non-collapsed decision rules can therefore produce reproducible corpora with per-decision provenance, while downstream effects differed between the evaluated domains.

**Index Terms**—pretraining data, dataset curation, data quality, auditability, decision provenance

## I. INTRODUCTION

Pretraining-data curation determines how much data a language model sees and which information remains available for learning. Large-corpus pipelines remove malformed content, apply rule-based or learned filters, and deduplicate at exact, fuzzy, or semantic levels [1], [3]–[5]. Scalable systems make these operators configurable [8], [10], but their final keep/drop output can hide why each item changed. This is important because an invalid record, a similarity match, a model judgment, and the loss of a rare content group are different kinds of evidence and should not authorize the same action.

Evidence-bound curation requires each proposed keep, remove, or restore action to pass an action-specific rule before it is applied. A similarity search may propose two items for comparison, but it cannot remove either item; removal requires a verified redundancy relation and a representative that remains in the output. Likewise, an uncertain Quality judgment is not a negative label, and Coverage may reverse a removal when support would disappear but may not create a new deletion. The evidence is therefore bound to the action it can justify.

In this paper, we introduce a three-stage framework that enforces this rule. We use Stage A to check explicit input validity and isolate failures from the usable pool. In Stage B, we verify redundancy relations and apply an abstention-aware, four-part Quality assessment. In Stage C, we check the combined result and restore support that earlier decisions would otherwise eliminate. We record the input fingerprint, rule version, reason, and representative or restoration link. Changing the text, evidence provider, or rule invalidates the earlier record instead of silently reusing it.

We evaluate the enforcement mechanism separately from the usefulness of its output. Runtime audits test whether unsupported actions are rejected and whether the output can be reproduced. We then use *natural-budget evaluation*: the frozen output is trained on every packed token it retains, without adding sampled data to match the Raw corpus. This preserves the practical consequence of curation—a smaller dataset produces fewer training updates—while benchmarks measure the resulting compression–capability trade-off. We report a three-seed code experiment and a three-seed math transfer.

These distinctions matter when the framework is reused. Detectors, rankers, retrieval methods, and thresholds may change, but the meaning of an accepted dataset action should not change silently with them. A usable curation record must therefore answer four questions: what evidence was observed, which action that evidence was allowed to trigger, which item or group supported the action, and whether the same decision can be replayed under the recorded versions. We treat these questions as runtime requirements rather than post-hoc annotations.

Figure 1 provides a visual summary of these stage-specific rules and the dataset actions they permit.

TABLE I

GUARANTEES STATED BY REPRESENTATIVE CURATION APPROACHES. THIS IS A DESIGN COMPARISON, NOT A BENCHMARK RESULT. “NOT STATED” MEANS THAT THE CITED DESIGN DOES NOT REQUIRE THE PROPERTY; AN IMPLEMENTATION COULD STILL BE CUSTOMIZED TO ADD IT.

| Approach                | Main decision record                 | Rule that permits an action                             | Representative or restoration safeguard                                   | Reproducibility check                                            |
|-------------------------|--------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------|
| QuRating [12]           | Four learned ratings                 | Ratings guide sampling or curriculum                    | Not stated                                                                | Not stated                                                       |
| D4 [13]                 | Cluster/prototype selection          | Selection algorithm defines the final set               | Diversity-aware prototypes, not a restore-only safeguard                  | Not stated                                                       |
| Data-Juicer [8]         | Operator recipe and trace            | Recipe order determines the final set                   | No system-wide safeguard stated                                           | Operator-level tracing                                           |
| NeMo Curator [10]       | Distributed filter and dedup outputs | Recipe-defined filtering and deduplication              | Algorithm-specific representative handling                                | Configurable execution artifacts                                 |
| Evidence-bound curation | Reason-coded decision log            | Isolate, remove, select, and restore use separate rules | Every removed duplicate keeps a representative; Coverage can only restore | Text, evidence, provider, and policy hashes are checked together |

Our contributions in this paper are summarized as follows.

- We define an evidence-bound decision model that gives Validity, Redundancy, Quality, and Coverage different permissions instead of collapsing their signals into one keep/drop score.
- We implement an A–B–C runtime that separates explicit isolation, verified representative-preserving removal, positive Quality selection, and restore-only Coverage.
- We bind every accepted action to content and policy identities, a reason code, and any required representative or restoration link, so that we can reject stale evidence and replay the final corpus.
- We evaluate one frozen policy on code and mathematics with three seeds, natural token budgets, and same-corpus Data-Juicer and NeMo Curator baselines, revealing both its compression and its domain-dependent downstream effects.

## II. RELATED WORK AND DESIGN GAP

### A. Corpus Construction and Quality Selection

C4, RefinedWeb, FineWeb, and Dolma show how extraction, content rules, and deduplication shape language-model corpora [1]–[4]. DataComp-LM compares recipes under controlled training [5]. QuRating learns four document ratings for sampling and curriculum construction, while D4 combines deduplication with embedding-based diversification [12], [13]. Yet a one-dimensional classifier may not predict filtering effects across domains or measure intrinsic quality [6], [7]. We instead govern how such evidence may change the output: one Core cannot compensate for another, and a Quality score cannot repair invalid input or remove the last supported member.

### B. Curation Systems

Data-Juicer provides composable operators, recipe exploration, and tracing [8]; NeMo Curator supplies distributed filters and exact, fuzzy, and semantic deduplication [10]; and Oasis combines rule filters, neural filtering, and corpus assessment [9]. They are mainly organized as processing pipelines or recipes. Ours is a complementary decision-control

layer that checks whether different observations are sufficient to keep, remove, or restore an item and then checks the combined result. NeMo does not require a top-level domain label; corpus compatibility is instead expressed through the selected operators and thresholds, including specialized code filters [10], [11]. Our selector consumes no declared domain, and the same frozen keep/remove/restore rules govern the reported code and math runs. The decision process therefore does not require a domain label; this is not a claim that the same Quality model works equally well in every domain. Table I compares stated guarantees rather than lists of operators. The distinction is whether a property remains mandatory when evidence providers and recipes are replaced. These approaches are therefore not interchangeable empirical baselines: QuRating and D4 define learned-rating or prototype-selection objectives, while Data-Juicer and NeMo Curator expose configurable execution systems. The same-corpus study includes both systems as executable baselines because one documented code- and math-compatible recipe from each system could be frozen and reproduced under the shared training protocol. It does not evaluate the full design space of QuRating, D4, Data-Juicer, or NeMo Curator.

### C. Why a Decision-Control Layer Is Distinct

An ordered filter pipeline computes a final keep/drop bit, but that bit does not identify which evidence was sufficient for the action. Evidence-bound curation adds a policy check before each dataset change and after policies are combined: similarity may retrieve a comparison, a verified Redundancy relation may remove a non-representative, and Coverage may restore but not exclude. Each accepted action is bound to the current text, evidence provider, and policy version, so changing any of them invalidates the earlier decision. These decision rules therefore stay explicit even when detectors, rankers, or retrieval systems are replaced.

### D. Deduplication and Failure Patterns

Deduplication reduces memorization and leakage [14], and The Stack already applies source-code near-deduplication [15].

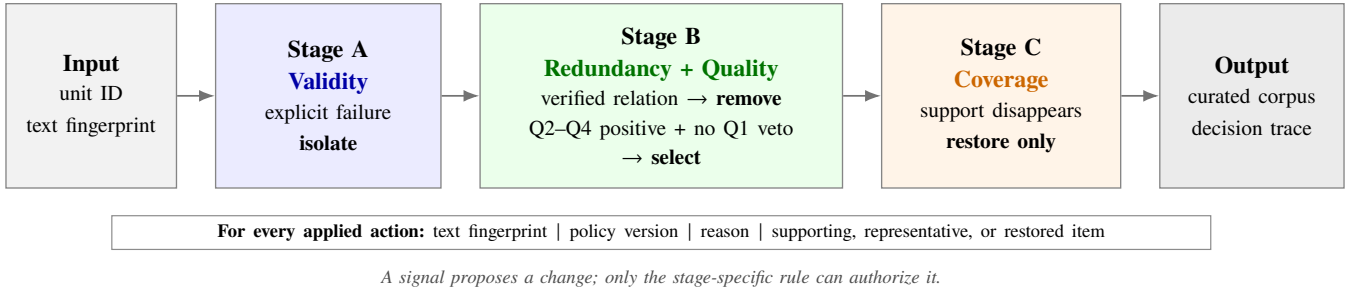


Fig. 1. Stage-specific evidence and allowed dataset actions. Retrieval and model judgments propose changes; the A–B–C rules decide whether to isolate, remove, select, or restore a training unit.

MinHash and LSH efficiently retrieve high-overlap candidates [16], but programs can share imports and structure while differing in behavior. For example, `return x >= 0` and `return x > 0` may collide without either representing the other. Conversely, uncertain partial fragments can all fail positive selection without being known incorrect. The rule set therefore requires relation verification for removal and permits Coverage to restore a representative when the combined local decisions remove every member of a group.

These cases expose a gap between finding a plausible signal and authorizing a dataset change. Candidate retrieval is deliberately recall-oriented, whereas removal needs a verified relation and a named survivor. A positive Quality judgment can justify keeping an item but cannot prove that a similar item is interchangeable; similarly, an uncertain judgment is not evidence that the item is invalid. The proposed runtime therefore records retrieval, verification, selection, and restoration as different operations. This separation lets the system reject an otherwise plausible action when its evidence does not carry the required permission.

### III. EVIDENCE-BOUND DECISION MODEL

#### A. What Each Core Is Allowed to Do

Let  $D$  be an audited input corpus and  $U(D)$  the deterministic training units created from it. A policy version  $\pi$  produces an output  $C_\pi \subseteq U(D)$  and a decision log  $T_\pi$ . For a unit  $x$ , an evidence producer emits  $e = (x, h_x, p, v, r)$ : unit identity, text hash, producer, version, and result. The observation remains descriptive until a policy permits a dataset change.

Consider two long functions retrieved as possible near duplicates. MinHash opens the comparison, but cannot remove either function. Redundancy may remove one only if an allowed relation is verified and a surviving representative is named. Quality selects a candidate only when Q2–Q4 provide positive support and no qualified Q1 failure vetoes that support. If those local decisions would erase the final member of a declared support group, Coverage restores one representative. The trace therefore explains both an exclusion and any later restoration.

Each Core has one noninterchangeable permission:

- 1) **Validity** isolates an explicitly defined input failure.

- 2) **Redundancy** removes only a verified non-representative whose representative remains in the output.
- 3) **Quality** positively selects a unit under conjunctive Q2–Q4 evidence, subject to a qualified Q1 failure veto; missing positive evidence is *not selected*, not “bad.”
- 4) **Coverage** restores support lost by combined proposals; it cannot originate an exclusion.

The separation is intentional because the four questions are not different estimates of one hidden score. A unit can be well formed but redundant, unique but unsupported by the Quality Ranker, or locally selectable while being the last member of a support group. Collapsing these cases into one score would let strong evidence on one axis compensate for a failure on another. It would also make the same numeric change mean isolation, representative removal, lack of positive selection, or restoration. The action-specific Core interface instead preserves both the reason for a decision and the action that reason is allowed to cause.

#### B. Rules for Dataset Changes

Every proposal  $p(x)$  that changes the corpus must name the current text hash  $h_x$ , its evidence  $e$ , and the versioned policy  $\pi$ :

$$p(x) \Rightarrow \text{valid}(\pi, e, h_x) \wedge \text{permitted}(\pi, e, x). \quad (1)$$

We enforce five testable properties at runtime. Every change names its evidence, policy, action, and reason. Every Redundancy removal leaves its named representative in the output. Coverage prevents the unexplained loss of every member of a support group. Downstream evaluation cannot feed back into selection. Finally, a rerun rejects reused evidence when the data, provider, model, policy, or threshold identity has changed.

Let  $V$  be Stage B-eligible units,  $P_\pi$  the reason-coded non-selection set, and  $\tilde{C}_\pi = V \setminus P_\pi$ . For every active support group  $g$ , Stage C requires a remaining member or a recorded policy exception  $a_g$ :

$$I_g(\tilde{C}_\pi) = \mathbf{1}[g \cap \tilde{C}_\pi \neq \emptyset \vee a_g = 1]. \quad (2)$$

If  $I_g = 0$ , Stage C records the IDs that must be kept,  $R_\pi$ , forms  $C_\pi = \tilde{C}_\pi \cup R_\pi$ , and reruns the checks. Missing or stale evidence rejects the proposed change rather than activating a looser fallback.

The trace distinguishes explicit rejection from absence of positive selection evidence. Benchmark scores, training loss, utility estimates, source reputation, target domain shares, retention fractions, and token budgets are not selector inputs. These rules govern what evidence may do; they do not claim that every detector is semantically correct.

Table II summarizes the unit-level contract. An absent unit must name a closed Validity reason, a surviving Redundancy representative, or a Quality selection decision; a restored unit names the Coverage group that required it. If text, evidence, or policy identity changes, a rerun rejects the inherited decision.

We design the resulting artifact to answer these questions without rerunning a benchmark or rescore the corpus. It records not only the curated corpus, but also a machine-checkable explanation of how that exact corpus was formed.

### C. How the Cores Work Together

We combine the Cores as an ordered sequence of dataset changes, not as majority voting among four classifiers. Stage A first determines which records and chunks are eligible for curation. Its isolation decisions are final because later evidence cannot repair missing or corrupt input. Stage B then creates a provisional keep set. Explicit evidence of non-training content may exclude a unit; a verified Redundancy relation may remove only the linked non-representative; and every remaining unit must satisfy the policy’s positive Quality-selection rule. A unit that lacks enough positive evidence is recorded as not selected, without being relabeled invalid or intrinsically low quality.

Stage C evaluates the provisional set as a whole. It receives support groups and the complete reason-coded proposal set, but no permission to create another exclusion. When Stage B has removed every member of a declared group without a recorded policy exception, Stage C restores one deterministic member. Commit is allowed only after all isolated items, surviving representatives, Quality decisions, and restoration links pass a second hash-and-identity check. Consequently, no Core can silently reinterpret another Core’s evidence: Quality cannot validate corrupt input, similarity cannot prove redundancy, and Coverage cannot declare a unit useful or delete it.

We evaluate and package one frozen beta policy rather than multiple strength profiles. The reported confirmatory runs materialize its Coverage decisions and pass the complete recheck, but do not claim production promotion. The policy receives no target corpus size, retention fraction, source prior, or domain quota. The final size is therefore an outcome of the recorded evidence rather than an input budget, and changing any rule or threshold creates a new policy identity rather than silently changing the meaning of the release.

## IV. CURATION PIPELINE

### A. Stage A: Explicit Validity Failures

The input adapter preserves declared metadata without inferring missing rights or source history. Stage A applies only content-preserving normalization and isolates observable

input failures: absent text, a declared text-format violation, unrecoverable corruption, or acquisition failure. Short code, partial files, unfamiliar formats, markup, multilingual text, and parser failure remain valid by default. Rights, PII, poisoning, and source-admission review are separate audit concerns; they are not silently converted into a text-quality score.

Released records are chunked deterministically into units of at most 6,000 characters. Every chunk retains its parent record ID and normalized text hash. Stage B first removes normalized exact duplicates, before candidate retrieval or model-assisted Quality is considered.

### B. Stage B: Verified Redundancy

Candidate retrieval combines exact text and formatting digests, exact containment-window digests, and MinHash LSH. The active retrieval settings are 5-token shingles, a 32-value signature, eight bands, and a 24-token retrieval minimum. The frozen policy considers near-duplicate pairs from 64 tokens with at most four changed tokens and 2%; units of at most 32 tokens are exact-only.

Retrieval only schedules comparisons. A removal proposal requires a verified relation allowed by policy: exact text, formatting equivalence, a near-duplicate that differs only within declared limits, prose reflow with unchanged tokens, exact token containment, or a declared equivalence verifier. Numeric, operator, negation, answer label, API-signature, code-identifier/keyword, and named-entity changes trigger substantive-difference protection. Pairwise matches are not chained into larger duplicate groups. The item that fully contains the other’s usable content is preferred as representative, with a stable ID tie break.

This split between retrieval and decision is essential to the paper’s claim. The completed run evaluated thousands of candidate pairs, yet every verified redundancy exclusion was justified by exact token containment rather than MinHash similarity alone (Section VII-A).

### C. Stage B: Explicit Non-Training Content Rules

Four explicit text rules detect a generated-and-do-not-edit artifact, a license-comment-only chunk, an empty HTML shell, or a chunk containing only a fixed vocabulary of web controls. Each requires positive in-text evidence. A generated source label alone, ordinary documentation, code with a license header, prose about cookies, and HTML with visible or executable content are protected negative cases. Exact sentences repeated at least three times within one chunk may be compacted when the sentence contains at least 12 lexical tokens. The transformation is recorded separately from whole-chunk selection and never supplies Redundancy or Quality evidence.

### D. Stage B: Quality Ranker and Positive Selection

A scalar Quality score can hide why a unit was selected and can let strength on one dimension compensate for a known failure on another. We therefore use four separate, non-compensating judgments; a strong result on one cannot cancel an explicit failure on another:

$$Q(x) = (Q_1, Q_2, Q_3, Q_4), \quad Q_i \in \{P, F, A\}, \quad (3)$$

TABLE II

DECISION TRACE FOR ONE TRAINING UNIT. EACH ROW NAMES THE RUNTIME QUESTION, THE ONLY ALLOWED EFFECT ON THE OUTPUT, AND THE RECORD NEEDED TO REPRODUCE THAT DECISION.

| Boundary           | Runtime question                                                   | Allowed effect                                                       | Record kept for reproduction                                                           |
|--------------------|--------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Bind               | Do unit, parent, text, provider, and policy identities match?      | Admit current evidence or reject it as stale                         | Stable UID, parent UID, normalized text hash, provider/model/config hashes             |
| Stage A            | Is an explicit input-rule failure observed?                        | Isolate; otherwise keep when uncertain                               | Validity reason code and normalized content identity                                   |
| Stage B retrieval  | Which units require comparison?                                    | Schedule a pair only; no output change                               | Candidate edge, retrieval method, pair identities                                      |
| Stage B Redundancy | Is an allowed removable relation verified?                         | Remove one non-representative only                                   | Relation type, evidence hash, representative UID                                       |
| Stage B Quality    | Do Q2–Q4 supply positive support without a qualified Q1 failure?   | Select or record that positive support was vetoed or not established | Q1–Q4 pass/fail/abstain results, confidence, out-of-distribution check, policy version |
| Stage C Coverage   | Did combined proposals leave a declared group without support?     | Restore a deterministic representative or reject the output          | Group/view identity, empty-group record, required-retain UID                           |
| Commit             | Do all identities, representatives, and support checks still hold? | Write the output once; otherwise abort                               | Ordered decision log, final unit list, corpus hash                                     |

TABLE III  
SEPARATE EVIDENCE HEADS USED BY THE QUALITY RANKER.

| Head | Positive question                                 | Explicit failure; otherwise abstain                 |
|------|---------------------------------------------------|-----------------------------------------------------|
| Q1   | Is there locally observable correctness evidence? | A locally checkable result is incorrect             |
| Q2   | Is the unit internally coherent?                  | The same context contains an explicit contradiction |
| Q3   | Does substantive content remain?                  | The unit contains only boilerplate or controls      |
| Q4   | Is at least one learnable relation recoverable?   | Only disconnected fragments remain                  |

where  $P$ ,  $F$ , and  $A$  denote pass, fail, and abstain. Table III states the question owned by each head and the evidence required for an explicit failure. These are specific, locally testable questions rather than claims of universal document quality.

The ranker is constructed in three steps. First, GPT-5.6 Luna supplies Q1–Q4 labels under a versioned Batch prompt. Second, four local classifier heads are fit over frozen Qwen3-Embedding-0.6B representations. Third, minimum confidence 0.8 and an out-of-distribution (OOD) cosine-similarity boundary of 0.3822 mark unsupported local predictions for fallback instead of forcing a label. External knowledge, undeclared execution assumptions, missing context, and unfamiliar notation therefore do not become automatic failures.

Q1 acts only as a qualified-failure veto, while positive membership support uses the conjunctive payload rule

$$\text{Select}(x) = \mathbf{1}[Q_1(x) \neq F \wedge Q_2(x) = P \wedge Q_3(x) = P \wedge Q_4(x) = P]. \quad (4)$$

If any local head abstains, is low-confidence, or is OOD, a frozen Luna observation applies the same questions. Conjunctive Q2–Q4 fallback support retains the chunk unless a qualified Q1 failure vetoes it; confirmed required-head failure or a completed fallback without positive support records it

as not selected. A missing observation or transport failure aborts materialization rather than changing membership. Thus unverifiable correctness is not treated as failure, but a locally checkable Q1 failure can veto selection; no weighted aggregate can compensate for a failed required head.

#### E. Stage C: Coverage Restoration

Coverage evaluates the combined Stage B result across overlapping views: redundancy families, deterministic content routes, language/script and format labels, residual links, and semantic support groups. The recorded semantic graph uses Qwen3-Embedding-0.6B as its primary provider and BGE-M3 as an independent audit view [25], [26]. Mutual eight-neighbor relations shared by both views define stable local support; neighborhoods found by only one provider are marked uncertain. Embedding similarity never authorizes an exclusion.

When a provisional output removes every member of a group without a recorded policy exception, Coverage chooses a deterministic representative, records why it is required, restores it, and performs a second complete check. Coverage does not maximize a diversity score, impose a quota, or infer that a rare group is useful. Its role is narrower: if all members of a declared group are removed without a recorded policy exception, the output is rejected.

#### F. Auditable Output

Each boundary writes unit and parent IDs, text and evidence hashes, policy/provider versions, decisions, reason codes, and any representative or restore link. The finalization step validates these identities, applies the A–B–C stages, reruns all decision checks after Coverage, and hashes the final corpus. Reports aggregate token and chunk impact by reason and use comparable Stage B-to-C denominators for composition changes.

Interrupted runs resume only when every bound identity matches. A stale or corrupt cache reduces usable evidence or aborts output construction; it cannot activate a permissive

fallback. The output corpus and its explanation are thus versioned together.

## V. IMPLEMENTATION

### A. Hash-Checked Execution

We separate policy specification, evidence artifacts, and final output generation. JSON configuration files bind the source manifest, tokenizer, chunking, model and provider identities, the active policy, and output locations. A runtime bridge rejects a resume if a bound identity or content hash changes. The release manifest binds one active policy and rejects legacy profile names. Table IV lists the versioned settings that materially affect the final corpus. Luna is an offline label and frozen fallback-evidence source, and Qwen3-4B-Base is used only for external training; neither is an undeclared selector feature.

The execution graph is partitionable at corpus scale. Stage A checks, per-unit Quality inference, and pairwise relation verification are keyed by stable unit or pair identities and require no shared mutable state. Exact digests and MinHash signatures can be built per shard before a keyed merge; Stage C is the only corpus-wide check, aggregating declared support groups before deterministic output construction. Decision traces are append-only and keyed by the same identities. This structure makes the decision layer compatible with distributed curation engines, although the present study does not report distributed throughput or storage benchmarks.

### B. Safe Output Construction

We build the output in one all-or-nothing step: if any required check fails, no final output is written. The steps are:

- 1) validate the source, tokenizer, schema, policy, and model identities;
- 2) keep or isolate records under Stage A and create deterministic chunk identities;
- 3) retrieve Redundancy candidates, verify allowed relations, apply closed non-training-content rules, and obtain version-bound Q1–Q4 evidence;
- 4) write the provisional keep set together with every decision and representative link;
- 5) run all Coverage views, restore required representatives, and repeat the complete decision check; and
- 6) hash the final corpus before exposing it to external training or evaluation.

An interrupted run resumes only when all bound identities still match. Missing or stale evidence reduces the set of permitted actions or aborts the run; it cannot activate a looser fallback. This lifecycle makes the curated corpus and its explanation one versioned artifact.

### C. User-Facing Inputs and Outputs

A deployment begins with JSONL records containing text and stable source-local identifiers; optional metadata is preserved but is not required to name a domain. A run configuration binds the frozen framework policy, tokenizer, normalization, frozen Quality Ranker, semantic providers, active

TABLE IV  
VERSIONED SETTINGS THAT AFFECT THE FINAL CORPUS.

| Component            | Setting                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------|
| Identity             | Stable UID, parent UID, normalized text/evidence SHA-256                                                             |
| Chunking             | Format preserving; maximum 6,000 characters                                                                          |
| Retrieval            | Exact/format/containment digests; 5-token shingles; 32-value MinHash; eight LSH bands                                |
| Redundancy           | Exact-only through 32 tokens; near duplicate $\geq 64$ tokens, $\leq 4$ changed tokens and $\leq 2\%$                |
| Non-training content | Four explicit in-text rules; repeated-sentence compaction requires $\geq 3$ occurrences and $\geq 12$ lexical tokens |
| Quality              | Qwen3-Embedding-0.6B local heads; confidence $\geq 0.8$ ; OOD cosine $\geq 0.3822$ ; Luna fallback                   |
| Quality selection    | Qualified Q1 failure veto; Q2 coherence $\wedge$ Q3 substantive payload $\wedge$ Q4 learnable relation required      |
| Coverage             | Qwen3 embedding primary, BGE-M3 audit view; mutual eight-neighbor support graph; veto only                           |
| Selector inputs      | Text and permitted evidence only; no utility, loss, benchmark, source, quota, or token budget                        |

policy files, and output location. Automatic mode computes the required embeddings and Coverage graph from all Stage B-eligible units. A caller may instead supply frozen artifacts with matching identities, but precomputed and automatically generated evidence cannot be mixed within one run.

The returned object is an audit bundle rather than only a filtered JSONL file. It contains normalized input, Stage A isolated and pass sets, Stage B selection and non-selection proposals, the final curated and not-selected corpora after Stage C, Redundancy representative links, Coverage restoration traces, per-reason token and chunk deltas, descriptive route and script composition reports, and a manifest of policy, provider, input, and output hashes. Thus a user can ask which evidence changed a unit’s keep/remove status, which representative justified a removal, whether Coverage reversed the proposal, and whether a later run used the same decision boundary. Composition reports remain descriptive and never force a target mixture.

### D. Scope of the Quality Model

The distilled ranker is a replaceable, versioned policy component rather than a universal quality classifier. Its encoder identity, four heads, OOD boundary, confidence thresholds, and protected-set checks are bound to the output trace. Replacing the teacher, encoder, labels, or curation policy invalidates that calibration and requires a new policy version; it cannot silently alter an existing corpus.

### E. Implementation Checks

Positive, false-positive, adversarial, and input-variation tests exercise each permitted action. Corpus reports check action-specific Quality decisions, representative survival, restore-only Stage C, and the complete post-Coverage recheck. Source, leakage, and packed-input reports bind hashes and exact token, block, and step counts. These checks show that the implemen-

TABLE V  
VALIDATION EVIDENCE AND ITS EXPLICIT BOUNDARY.

| Check             | Observation                                                                                                                | Unsupported inference                       |
|-------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Core behavior     | 39 constructed positive and negative cases exercise permitted and forbidden actions                                        | Accuracy outside specified fixtures         |
| Quality selection | Q2-Q4 conjunctive rule, qualified Q1 veto, and fallback identity are recorded; Code/Math not-selected counts are 816/1,223 | Intrinsic or universal document quality     |
| Redundancy        | Code/Math verified removals are 3/6; each names a surviving representative                                                 | Recall over all duplicate relations         |
| Coverage          | Code/Math restore 104/143 units; both complete rechecks pass                                                               | Semantic correctness of every support group |

tation follows the stated rules; they do not establish semantic accuracy or downstream benefit.

Table V separates tested behavior from semantic claims. The protected Quality fixtures are UID- and text-hash-disjoint from calibration. Their outcomes are teacher-relative, not human ground truth. Likewise, Coverage support groups are reproducible graph artifacts, not estimates of semantic precision.

## VI. EXPERIMENTAL SETUP

We design the experiments to test two distinct questions. Runtime evidence checks whether the implementation allows only permitted dataset changes; continued pretraining and benchmarks characterize the completed corpora after their contents are fixed.

### A. Corpora and Benchmark-Overlap Checks

The code Raw arm contains 4,228 records from bigcode/the-stack-dedup and 662 records from pinned snapshots of eight open-source repositories. A pre-curation audit removes 12 exact or 16-token-shingle benchmark matches. This establishes the declared text-matching boundary, not absence of semantic or base-model contamination. Quality calibration and protected samples are UID- and text-hash-disjoint. The full run uses the fixed local ranker and frozen Luna fallback observations, without live teacher calls, benchmark, utility, source-score, or token-budget inputs.

The exploratory math arm is a separately sampled OpenWebMath corpus [21]. A pre-curation audit removes one full-MATH containment match from 3,232 records and finds no remaining MATH-500 match under the declared text-matching test [22], [23]. The same frozen framework policy is applied to code and math without a declared domain label or domain-specific retuning.

For direct external comparisons, the audited code and math inputs are also processed by frozen Data-Juicer and NeMo Curator 1.3.0 recipes. The Data-Juicer code recipe adapts the published Stack refinement recipe: normalization and cleanup precede length, repetition, and alphanumeric filters and document-level SimHash deduplication. Its math recipe similarly adapts the published arXiv refinement recipe, retaining math-compatible character, length, repetition, and perplexity filters followed by SimHash deduplication [8]. The NeMo

code recipe applies the documented Python comment-to-code, line-count, alphabetic-content, and XML-header filters. The math-compatible recipe uses length, boilerplate, repeated-line, repeated-paragraph, and repeating- $n$ -gram filters while excluding text heuristics that would penalize mathematical notation. Both then apply MD5 exact and MinHash-LSH fuzzy deduplication [10], [11]; semantic deduplication and target token budgets are disabled. Each system arm represents one declared recipe, not the full configuration space of either system.

### B. Training and Evaluation Protocol

We continue Qwen3-4B-Base [24] on audited Raw, the final framework output, Data-Juicer, and NeMo outputs. Each arm uses its complete natural token stream; there is no equal-token resampling, target retention fraction, or maximum-token cap. The final framework code experiment and the math transfer use seeds 101, 202, and 303. All arms share the same model revision, tokenizer, one-pass packing, sequence length 2,048, QLoRA configuration, and optimization recipe; only the selected data and resulting number of steps differ. We record the complete training, decoding, task-snapshot, and evaluator identities by hash in the reproducibility package. The hash-checked artifact is available at [https://github.com/LeeKeonSoo/UNLV-Research/blob/main/paper/Evidence\\_Bound\\_Curation\\_Reproducibility.zip](https://github.com/LeeKeonSoo/UNLV-Research/blob/main/paper/Evidence_Bound_Curation_Reproducibility.zip).

We report deterministic pass@1 separately for HumanEval+, MBPP+, BigCodeBench Complete, CRUXEval-I/O, and DS-1000; every arm, seed, and task is retained rather than collapsed into a single aggregate score [17]–[20].

We reuse the same recipe in the math experiment and evaluate Raw, the final framework output, Data-Juicer, and NeMo Curator with seeds 101, 202, and 303 on frozen GSM8K and MATH-500 snapshots. Math remains an exploratory transfer rather than a confirmatory domain-general endpoint.

GSM8K strict and flexible extraction can disagree when a response continues into another generated problem. We retain both official scores and report one identically applied normalized rescore as a post-hoc sensitivity audit; it does not replace either official endpoint.

## VII. RESULTS

We first report the frozen corpora and decision path. Because Raw units are records and curated units are chunks, corpus comparisons use tokenizer counts rather than unit counts. The code corpus changes from 6,984,438 to 6,249,197 tokenizer tokens (89.47% retained), and math changes from 6,984,004 to 5,780,762 (82.77% retained). Complete optimizer groups contain 6,242,304 and 5,767,168 training tokens, respectively.

### A. Decision and Output Audit

Figure 2 summarizes packed-token retention and each method’s mean score change from Raw, while Table VI preserves the absolute scores. Every arm consumes its complete natural budget. Figure 3 shows how the same frozen policy acts

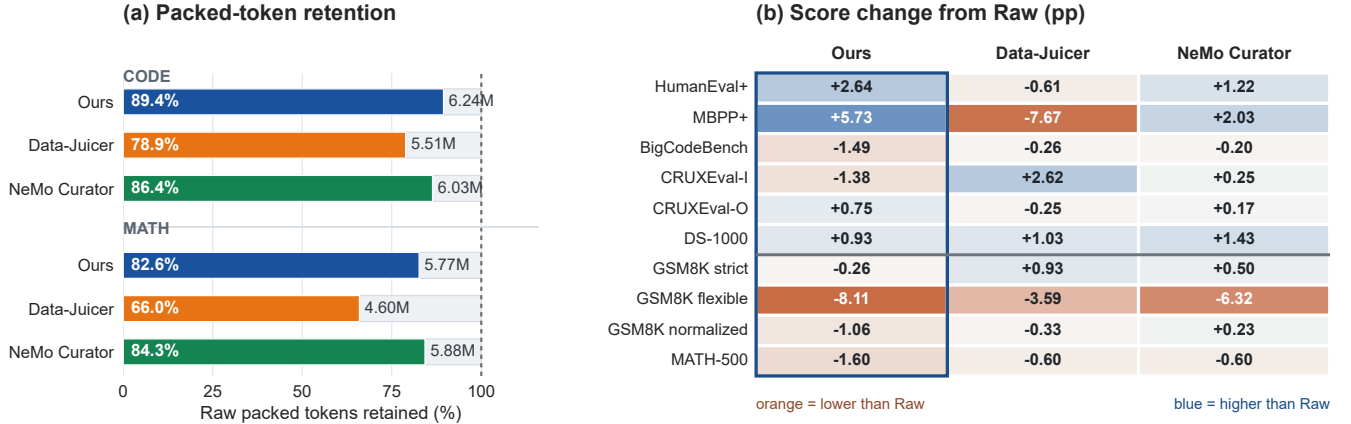


Fig. 2. Compression and downstream trade-offs under natural-budget evaluation. (a) Packed-token retention relative to Raw, with exact retained token counts. (b) Three-seed mean score change from Raw; positive values indicate improvement. Table VI reports the corresponding absolute scores.

TABLE VI

THREE-SEED DOWNSTREAM SCORES AS MEAN  $\pm$  SAMPLE STANDARD DEVIATION. BASE IS A SINGLE DETERMINISTIC NO-UPDATE REFERENCE; ALL OTHER ARMS USE THEIR COMPLETE NATURAL TOKEN STREAM.

| Benchmark              | Base  | Raw              | Ours             | Data-Juicer      | NeMo Curator     |
|------------------------|-------|------------------|------------------|------------------|------------------|
| <b>Code Benchmarks</b> |       |                  |                  |                  |                  |
| HumanEval+             | 31.10 | 28.86 $\pm$ 1.53 | 31.50 $\pm$ 0.93 | 28.25 $\pm$ 0.70 | 30.08 $\pm$ 2.54 |
| MBPP+                  | 58.47 | 48.94 $\pm$ 3.05 | 54.67 $\pm$ 0.85 | 41.27 $\pm$ 1.32 | 50.97 $\pm$ 0.85 |
| BigCodeBench           | 39.65 | 40.61 $\pm$ 0.63 | 39.12 $\pm$ 0.53 | 40.35 $\pm$ 0.40 | 40.41 $\pm$ 0.10 |
| CRUXEval-I             | 3.50  | 7.17 $\pm$ 0.69  | 5.79 $\pm$ 1.28  | 9.79 $\pm$ 0.62  | 7.42 $\pm$ 0.76  |
| CRUXEval-O             | 2.13  | 2.08 $\pm$ 0.75  | 2.83 $\pm$ 0.29  | 1.83 $\pm$ 0.26  | 2.25 $\pm$ 0.33  |
| DS-1000                | 28.00 | 33.20 $\pm$ 0.36 | 34.13 $\pm$ 0.46 | 34.23 $\pm$ 0.98 | 34.63 $\pm$ 0.72 |
| <b>Math Benchmarks</b> |       |                  |                  |                  |                  |
| GSM8K strict           | 75.13 | 77.79 $\pm$ 1.32 | 77.53 $\pm$ 0.39 | 78.72 $\pm$ 0.29 | 78.29 $\pm$ 0.84 |
| GSM8K flexible         | 57.54 | 71.19 $\pm$ 1.49 | 63.08 $\pm$ 2.97 | 67.60 $\pm$ 0.94 | 64.87 $\pm$ 2.90 |
| GSM8K normalized       | 76.42 | 80.11 $\pm$ 0.68 | 79.05 $\pm$ 0.42 | 79.78 $\pm$ 0.16 | 80.34 $\pm$ 0.58 |
| MATH-500               | 3.80  | 8.67 $\pm$ 0.31  | 7.07 $\pm$ 0.64  | 8.07 $\pm$ 0.46  | 8.07 $\pm$ 0.70  |

on both domains. Candidate retrieval alone removes nothing, every Redundancy exclusion names a survivor, and Coverage restores unsupported groups before the final recheck.

### B. Composition Effects

Source identity and target domain shares are not selector inputs. The output bundle nevertheless reports content route, document format, language/script, and document-function shares before and after curation. These deterministic labels explain composition shifts; they are neither semantic ground truth nor target proportions and never force a quota.

### C. Code Results

We completed the three-seed framework evaluation under the frozen protocol. HumanEval+ is complete at 31.50%, compared with 31.10% for Base and 28.86% for Raw. MBPP+ reaches 54.67%, below Base (58.47%) but above Raw (48.94%), Data-Juicer (41.27%), and NeMo Curator (50.97%). CRUXEval-I/O reach 5.79%/2.83%; the input score trails Raw

and both curation baselines, whereas the output score exceeds all four references. BigCodeBench reaches 39.12%, below all four references, whereas DS-1000 reaches 34.13%, above Base and Raw but below Data-Juicer and NeMo Curator. Overall, the framework exceeds both same-corpus curation baselines on three of six Code benchmarks and exceeds Base on four of six.

### D. Exploratory Math Results

The same frozen policy retains 82.77% of the math tokenizer stream without a domain label or math-specific threshold. Across three seeds, it scores 77.53% on GSM8K strict, 63.08% on GSM8K flexible, 79.05% under the normalized sensitivity rescore, and 7.07% on MATH-500. All four values exceed Base but trail Raw and both executable curation baselines. This transfer shows that the decision process executes consistently outside code, while the downstream result does not establish domain-general effectiveness.



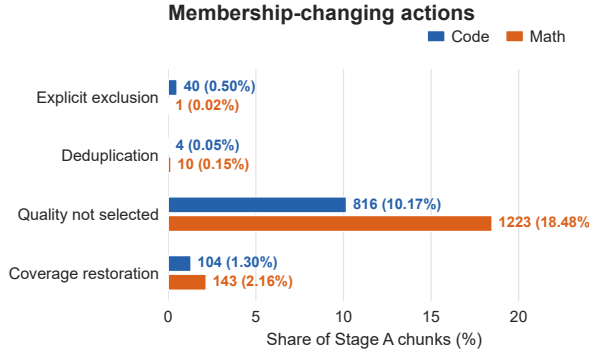


Fig. 3. Membership-changing actions in the frozen policy, normalized by Stage A chunks; labels give exact counts and percentages. Stage A contains 8,026 code and 6,619 math chunks, and the final outputs contain 7,270 and 5,528. Both rechecks leave zero unsupported groups. Coverage values are restorations, not exclusions. Deduplication combines 1/4 normalized exact duplicates and 3/6 verified representative-preserving removals in Code/Math; repeated-sentence transformations are omitted because they do not change membership.

## VIII. DISCUSSION

### A. What the Runtime Establishes

Runtime audits and downstream training answer different questions. The former show that retrieval alone caused no deletion, all verified removals retained a representative, and 104/143 Coverage restores prevented the unsupported loss of every item in a Code/Math support group. The latter evaluates each completed corpus under its natural budget. A benchmark gain cannot validate every local decision, and a regression does not invalidate the trace; it prevents us from claiming that the frozen policy is effective.

### B. Tracing the Mixed Results

We use the decision log to identify the dominant intervention. Explicit structural rules exclude 40/1 chunks, normalized exact deduplication removes 1/4, and verified Redundancy removes 3/6 in code/math, whereas Quality records 816/1,223 not-selected chunks before Coverage restores 104/143. Repeated-sentence compaction transforms 222/443 chunks without changing their membership. Coverage prevents complete group loss, not utility loss within a surviving group, so the trace supports diagnosis without claiming a causal mechanism. The local heads resolve 1,807/7,982 Code decisions but only 48/6,608 Math decisions; frozen Luna fallback observations supply the remaining 77.36%/99.27%. The Math transfer therefore primarily evaluates the frozen teacher-backed fallback rather than autonomous local-ranker transfer.

### C. Relationship to Existing Systems

QuRating strengthens evidence production, while NeMo Curator and Data-Juicer provide operator libraries and scalable execution. Evidence-bound curation is a complementary decision layer: retrieval proposes, a verified relation may remove,

positive Quality evidence may select, and Coverage may only restore. Hash and identity checks enforce these limits while allowing the ranker, retriever, support graph, or execution engine to be replaced.

### D. Interpreting the Frozen Comparison

The completed comparison characterizes the compression-capability trade-off of the single framework output relative to Raw and the two executable systems. Compression alone does not establish success, and the task-level outcomes remain mixed. Our distinct claim is the reproducible decision path; downstream scores bound how strongly the particular frozen policy can be recommended.

## IX. LIMITATIONS

We complement existing curation operators rather than claiming that general curation systems are absent. Our contribution is the enforceable and inspectable connection between different forms of evidence and allowed dataset changes; the effectiveness of any frozen policy still requires empirical evaluation.

The evaluation compares complete curation systems, not rule-level causal effects. Raw’s Stack component was already deduplicated, limiting observable Redundancy removal, and retention may reflect length, style, familiarity, or usable content. Normalized GSM8K is a post-hoc sensitivity audit. Quality calibration uses synthetic fixtures and teacher labels; the stored observations are versioned and reusable, but they cannot be independently regenerated without the same proprietary teacher. Coverage graphs define reproducible support candidates rather than semantic ground truth. Their reported materializations and complete rechecks are limited to the confirmatory code and math runs and do not constitute production promotion. Linking available repository rights metadata to every record also remains incomplete for a self-contained corpus release.

We evaluate at a smaller scale than production pretraining: two roughly 6.98M-token corpora and QLoRA continued pretraining. The final code and trained math arms have three seeds. Distributed throughput, wall time, peak memory, and energy are not benchmarked. General-text and multilingual effects remain unevaluated, and each external-system comparison covers one frozen recipe. The keep/remove/restore rules are evaluated only under the reported corpora and training protocol. The method-level approaches in Table I are not same-corpus trained baselines. Claims therefore apply to the reported corpora, policies, model, recipes, and scorers rather than universal downstream effectiveness.

## X. CONCLUSION

Evidence-bound curation makes keep, remove, and restore decisions auditable by giving Validity, Redundancy, Quality, and Coverage separate actions. In the final code and math runs, similarity retrieval alone never removed data, every Redundancy removal named a surviving representative, and Coverage restored support before output materialization. The

same single policy retains 89.47% of code and 82.77% of math tokenizer tokens and passes the complete runtime recheck without using target size or downstream utility.

The completed evaluations show mixed downstream effects: the frozen policy exceeds both curation baselines on three of six Code benchmarks but trails them on all four reported Math scores. The frozen beta decision layer does not require a declared domain and keeps each corpus change tied to reproducible evidence. The empirical claim therefore remains limited to the completed code and math protocols.

## REFERENCES

- [1] C. Raffel et al., “Exploring the limits of transfer learning with a unified text-to-text transformer,” *J. Mach. Learn. Res.*, vol. 21, no. 140, pp. 1–67, 2020.
- [2] G. Penedo et al., “The RefinedWeb dataset for Falcon LLM: Outperforming curated corpora with web data only,” in *Proc. NeurIPS Datasets and Benchmarks Track*, vol. 36, 2023, doi: 10.52202/075280-3464.
- [3] G. Penedo et al., “The FineWeb datasets: Decanting the web for the finest text data at scale,” in *Proc. NeurIPS Datasets and Benchmarks Track*, vol. 37, 2024, doi: 10.52202/079017-0970.
- [4] L. Soldaini et al., “Dolma: An open corpus of three trillion tokens for language model pretraining research,” in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (ACL)*, pp. 15725–15788, 2024, doi: 10.18653/v1/2024.acl-long.840.
- [5] J. Li et al., “DataComp-LM: In search of the next generation of training sets for language models,” in *Proc. NeurIPS Datasets and Benchmarks Track*, 2024.
- [6] S. Longpre et al., “A pretrainer’s guide to training data: Measuring the effects of data age, domain coverage, quality, & toxicity,” in *Proc. NAACL-HLT*, pp. 3245–3276, 2024, doi: 10.18653/v1/2024.naacl-long.179.
- [7] T. Nait Saada, L. Bethune, M. Klein, D. Grangier, M. Cuturi, and P. Abblin, “The data-quality illusion: Rethinking classifier-based quality filtering for LLM pretraining,” arXiv:2510.00866, 2025.
- [8] D. Chen et al., “Data-Juicer: A one-stop data processing system for large language models,” in *Companion Proc. 2024 Int. Conf. Manage. Data (SIGMOD-Companion)*, pp. 120–134, 2024, doi: 10.1145/3626246.3653385.
- [9] T. Zhou, Y. Chen, P. Cao, K. Liu, J. Zhao, and S. Liu, “Oasis: Data curation and assessment system for pretraining of large language models,” in *Proc. 33rd Int. Joint Conf. Artif. Intell. (IJCAI), Demo Track*, pp. 8855–8859, 2024, doi: 10.24963/ijcai.2024/1048.
- [10] NVIDIA, “Overview of NeMo Curator,” *NeMo Curator Documentation*. [Online]. Available: <https://docs.nvidia.com/nemo/curator/main/about>. Accessed: Aug. 20, 2026.
- [11] NVIDIA, “Quality assessment & filtering,” *NeMo Curator Documentation*. [Online]. Available: <https://docs.nvidia.com/nemo/curator/latest/curate-text/process-data/quality-assessment>. Accessed: Aug. 20, 2026.
- [12] A. Wettig, A. Gupta, S. Malik, and D. Chen, “QuRating: Selecting high-quality data for training language models,” in *Proc. ICML*, 2024.
- [13] K. Tirumala, D. Simig, A. Aghajanyan, and A. S. Morcos, “D4: Improving LLM pretraining via document de-duplication and diversification,” in *Proc. NeurIPS Datasets and Benchmarks Track*, vol. 36, 2023.
- [14] K. Lee, D. Ippolito, A. Nystrom, C. Zhang, D. Eck, C. Callison-Burch, and N. Carlini, “Deduplicating training data makes language models better,” in *Proc. ACL*, pp. 8424–8445, 2022.
- [15] D. Kocetkov et al., “The Stack: 3 TB of permissively licensed source code,” *Trans. Mach. Learn. Res.*, 2023.
- [16] A. Z. Broder, “On the resemblance and containment of documents,” in *Proc. Compression and Complexity of Sequences*, pp. 21–29, 1997.
- [17] J. Liu, C. S. Xia, Y. Wang, and L. Zhang, “Is your code generated by ChatGPT really correct? Rigorous evaluation of large language models for code generation,” in *Proc. NeurIPS*, 2023.
- [18] T. Y. Zhuo et al., “BigCodeBench: Benchmarking code generation with diverse function calls and complex instructions,” in *Proc. ICLR*, 2025.
- [19] A. Gu, B. Rozière, H. Leather, A. Solar-Lezama, G. Synnaeve, and S. I. Wang, “CRUXEval: A benchmark for code reasoning, understanding and execution,” in *Proc. 41st Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 235, pp. 16568–16621, 2024.
- [20] Y. Lai et al., “DS-1000: A natural and reliable benchmark for data science code generation,” in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 202, pp. 18319–18345, 2023.
- [21] K. Paster, M. Dos Santos, Z. Azerbayev, and J. Ba, “OpenWebMath: An open dataset of high-quality mathematical web text,” arXiv:2310.06786, 2023.
- [22] K. Cobbe et al., “Training verifiers to solve math word problems,” arXiv:2110.14168, 2021.
- [23] D. Hendrycks, C. Burns, S. Kadavath, A. Arora, S. Basart, E. Tang, D. Song, and J. Steinhardt, “Measuring mathematical problem solving with the MATH dataset,” in *Proc. NeurIPS Datasets and Benchmarks Track*, 2021.
- [24] A. Yang et al., “Qwen3 technical report,” arXiv:2505.09388, 2025.
- [25] Y. Zhang et al., “Qwen3 Embedding: Advancing text embedding and reranking through foundation models,” arXiv:2506.05176, 2025.
- [26] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” in *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 2318–2335, 2024, doi: 10.18653/v1/2024.findings-acl.137.